

SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

Milestone 3 Report

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Program: Infosys Virtual Internship

Project: SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

Framework: YOLOv8 (Ultralytics)

Hardware Used: NVIDIA GTX 1650 Ti GPU (4GB)

Software: Python, PyTorch, Ultralytics YOLOv8, OpenCV, VS Code

Milestone 3 Objectives

The main objectives of Milestone 3 were:

- Deploy the trained YOLOv8 model into a real-time system
 - Implement live video processing using OpenCV
 - Detect safety equipment and violations in real time
 - Generate alerts for missing safety equipment
 - Test the system under different real-world scenarios
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Real-Time Detection System Implementation

Objective

The goal of Week 5 was to convert the trained YOLOv8 model into a real-time safety monitoring system.

Real-Time Video Processing

- The webcam was accessed using OpenCV
- Frames were captured continuously in real time
- Each frame was processed individually

Implementation

```
cap = cv2.VideoCapture(1)
```

Object Detection using YOLOv8

- Each frame was passed to the trained YOLO model
- The model detects multiple classes including:
- Person
- Hardhat
- Mask
- Safety Vest
- Machinery
- Vehicle
- Safety Cone

Detection Command

```
results = model(frame, conf=0.5)
```

Detection Output

The model produces:

- Bounding boxes
- Class labels
- Confidence scores

These outputs are visualized on the screen in real time.

Safety Violation Detection Logic

Objective

The objective of Week 6 was to implement person-wise safety violation detection.

Person-wise Analysis

- Detected objects are divided into:
 - Persons
 - Safety equipment
 - For each detected person:
The system checks whether safety equipment is present inside the person's bounding box
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Safety Rules Implemented

The following rules were applied:

- Missing helmet → Helmet Missing
 - Missing mask → Mask Missing
 - Missing safety vest → Safety Vest Missing
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Violation Detection Logic

- If safety equipment is not found inside the person's bounding box, it is considered a violation
 - Violations are stored in a list
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Alert Generation

- Alerts are displayed on the video screen in real time
- Alerts are also printed in the console

Example Output

ALERT: Helmet Missing

ALERT: Safety Vest Missing

Visualization and UI

- YOLO automatically displays bounding boxes and labels
 - Alerts are shown as text on the video frame
 - Multi-line alerts are implemented for better readability
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Testing and Validation

Testing Approach

The system was tested using:

- Real-time webcam input
 - AI-generated test images displayed through a mobile device
 - Pre-recorded video inputs for realistic scenario testing
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Real-Time Video Testing

- The trained model was tested on pre-recorded videos of construction site environments
- Video frames were processed similarly to live webcam input
- The system successfully detected objects and safety violations in video streams

Key Observations from Video Testing

- The model detected multiple workers in dynamic scenes
 - Safety violations such as missing helmet and safety vest were correctly identified
 - The system maintained smooth performance during video playback
 - Real-time alerts were displayed accurately over video frames
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Test Scenarios

The following scenarios were tested:

- Worker wearing full safety equipment
- Worker missing helmet
- Worker missing mask

- Worker missing safety vest
 - Multiple workers with mixed safety conditions
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Results

- The system successfully detected safety violations in real time
 - Alerts were generated correctly for missing equipment
 - The system performed smoothly with both webcam and video inputs
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Observations

- The model performed well in controlled lighting conditions
 - Detection accuracy decreased slightly in complex scenes
 - Small objects and occlusions affected detection performance
 - Person-wise safety mapping improved detection accuracy
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Challenges Faced

- Incorrect class label mapping in dataset
 - Case sensitivity issues in class names
 - Misclassification between safety and non-safety classes
 - Performance lag when adding advanced features
 - Difficulty in handling multiple persons initially
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Solutions Implemented

- Corrected dataset labels and class mapping
 - Normalized class names to avoid case issues
 - Improved logic for person-wise safety detection
 - Optimized code for real-time performance
 - Simplified alert system to reduce lag
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Performance Summary

Feature	Status
Real-time detection	Completed
YOLO object detection	Completed
Person-wise safety logic	Completed
Alert system	Completed
UI visualization	Completed
Testing	Completed

Future Improvements

- Add sound alerts using non-blocking methods
 - Highlight unsafe persons with colored bounding boxes
 - Improve dataset for better accuracy
 - Implement person tracking (ID-based monitoring)
 - Deploy system on edge devices (CCTV / IoT systems)
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Milestone 3 Results

Key outcomes achieved during Milestone 3:

- Real-time YOLOv8 detection system successfully implemented
 - Safety violation detection logic developed
 - Alert system integrated with real-time video
 - System tested under multiple scenarios including video inputs
 - Smooth and efficient performance achieved
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Milestone 3 Conclusion

Milestone 3 focused on deploying the trained YOLOv8 model into a real-time safety monitoring system. The system captures live video using OpenCV, performs object detection, and identifies safety violations such as missing helmet, mask, and safety vest.

A person-wise detection approach was implemented to improve accuracy, ensuring that safety equipment is correctly associated with each individual. The system was successfully tested using both live webcam input and recorded video streams, demonstrating its robustness in dynamic environments.

The system demonstrates the effective application of computer vision in enhancing workplace safety and provides a strong foundation for future improvements.